

Two mechanisms of bacterial killing

- ❑ **Oxygen-dependent mechanism**
reactive oxygen species – superoxide anion, hydroxyl ion, hydroperoxide
- ❑ **Oxygen Independent Mechanisms**
– using the content of granules (lysozyme, proteins influencing bacterial cell wall)

 ppt Online

[Visit](#)

Inflammaione. (Subject 4) -
online presentation

Images may be subject to copyright. [Learn More](#)

Related images

S. No.	Mechanism Involved	Role
1.	Bactericidal permeability increasing protein (BPI)	Phospholipase activation thereby phospholipid membrane degradation
2.	Lysozyme	Hydrolysis of muramic acid -N-acetylglucosamine bond present in the glycopeptide coat of bacteria
3.	Lactoferrin	Iron binding protein prevent bacterial proliferation and invasion
4.	Major basic protein (MBP)	Present in eosinophilic granules having cytotoxic action to parasites and possess little bactericidal action
5.	Defensins	Arginine rich granules forms hole in bacterial membranes

Anatomical Barriers - Mechanical Factors

System or Organ	Cell type	Mechanism
Skin	Squamous epithelium	Physical barrier Desquamation
Mucous Membranes	Non-ciliated epithelium (e.g. GI tract)	Peristalsis
	Ciliated epithelium (e.g. respiratory tract)	Mucociliary elevator
	Epithelium (e.g. nasopharynx)	Flushing action of tears, saliva, mucus, urine

Innate important lecture
[slideshare.net](#)

Mechanisms involved in oxyg...
[researchgate.net](#)

Oxygen independent mechanism

- Lysozymes and lactoferrins as well as certain cationic proteins act as bacteriostatic substances, which are oxygen independent and function under anaerobic conditions.
- Finally hydrolytic enzymes digest the killed organisms and the degraded products are released to the exterior.

Immunity in some important bacterial infections

Infection	pathogenesis	major defence mechanisms
Corynebacterium diphtheriae	non-invasive pharyngitis – toxin	neutralizing antibody
Vibrio cholerae	non-invasive enteritis – toxin	neutralizing and adhesion blocking antibodies
Neisseria meningitidis (Gram-negative)	nasopharynx → bacteraemia → meningitis → endotoxaemia	killed by antibody and lytic complement; opsonized and phagocytosed




2.CLASSIFICATION

Two types of tumor antigens have been identified on tumor cells:

1. **Tumor-Specific Transplantation Antigens (TSTAs)**
E.g. : Abnormal products of ras and p53 genes
2. **Tumor-Associated Transplantation Antigens (TATAs)**
E.g. : Epidermal growth factor (EGF) like p97



SlideShare

Visit

Tumor antigen

Images may be subject to copyright. [Learn More](#)

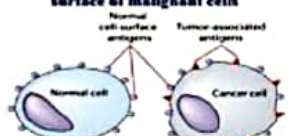
Related images



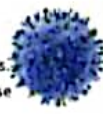
3.PROPERTIES

3.2. Tumor-associated transplantation antigens (TATAs)

Tumor associated antigens on surface of malignant cells


Tumor antigen
slideshare.net


1.TUMOUR ANTIGEN



- Tumor antigen is an antigenic substance produced in tumor cells. i.e., it triggers an immune response in the host.
- These antigens are not membrane proteins but are derivatives of cytosolic proteins. i.e., peptides of cytosolic proteins

Tumor antigen
slideshare.net


2.CLASSIFICATION

Two proteins- red and blue

Two genes- red and blue

Introduction

- Tumor immunology is mainly to study the immunogenicity of tumor and the mechanism of immune response to tumor, to demonstrate the relationship between the status of immune system and

Classification of Autoimmune Diseases

2. Localised autoimmune diseases or Organ specific autoimmune diseases

- A particular organ is affected due to auto Abs.
- For example:
 - **Thyroiditis** (attacks the thyroid)
 - **Multiple sclerosis** (attacks myelin coating of nerve axons)
 - **Myasthenia gravis** (attacks nerve-muscle junction)
 - **Juvenile diabetes** or Type I DM (attacks insulin-producing cells)



Auto immune diseases

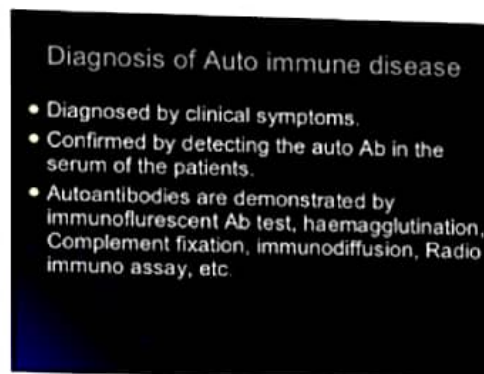
Visit

Images may be subject to copyright. [Learn More](#)

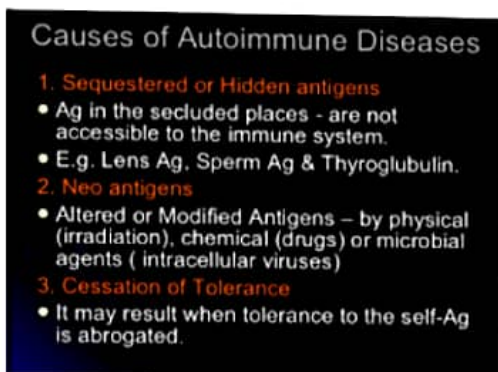
Related images



Auto Immune diseases
slideshare.net



Auto Immune diseases
slideshare.net



Autoimmunity Classification

Can be classified into clusters that are either
organ-specific or systemic

Organ-specific autoimmune diseases	Systemic autoimmune diseases
Type I diabetes mellitus	Rheumatoid arthritis
Goodpasture's syndrome	Scleroderma
Multiple sclerosis	Systemic lupus erythematosus Primary Sjögren's syndrome Polymyositis
Graves' disease Hashimoto's thyroiditis Autoimmune pernicious anemia Autoimmune Addison's disease Vitiligo Myasthenia gravis	

Figure 13-1 Immunobiology, 6/e. (© Garland Science 2005)



SlideShare

Autoimmunity

Visit

Images may be subject to copyright. [Learn More](#)

Related images

Some common autoimmune diseases classified by immunopathogenic mechanism		
Syndrome	Autoantigen	Consequence
Type II antibody against cell-surface or matrix antigens		
Autoimmune hemolytic anemia	Rh blood group antigens, I antigen	Destruction of red blood cells by complement and Fc γ phagocytes, anemia
Autoimmune thrombocytopenic purpura	Platelet integrin glycoproteins	Abnormal bleeding
Goodpasture's syndrome	Basement membrane domains of basement membrane collagen type IV	Glomerulonephritis, pulmonary hemorrhage
Psoriasis vulgaris	Epidermal keratin	Blistering of skin
Acute rheumatic fever	Streptococcal cell wall antigens. Antibodies cross-react with cardiac muscle	Arthritis, myocarditis, later scarring of heart valves

Autoimmunity - Microbiology ...
slideshare.net

Autoimmunity : General principles and observations

- Autoimmunity results from a failure or breakdown of the mechanisms normally responsible for maintaining self-tolerance in B cells, T cells, or both.
- The major factors that contribute to the development of autoimmunity are genetic susceptibility and environmental triggers, such as infections.
- Autoimmune diseases may be either systemic or organ specific.
- Various effector mechanisms are responsible for tissue injury in different autoimmune diseases.
- Epitope spreading: Autoimmune reactions initiated against one self antigen that injure tissues may result in the release and alterations of other tissue antigens, activation of lymphocytes specific for these other antigens, and exacerbation of the disease.

Autoimmunity
slideshare.net

Classification of Autoimmune Diseases

2. Localised autoimmune diseases or Organ specific autoimmune diseases

- A particular organ is affected due to auto Abs.
- For example:
 - Thyroiditis (attacks the thyroid)
 - Multiple sclerosis (attacks myelin coating of nerve axons)
 - Myasthenia gravis (attacks nerve-muscle junction)
 - Juvenile diabetes or Type I DM (attacks insulin-producing cells)

Auto immune diseases
slideshare.net

Types of autoimmune diseases

- Autoimmune diseases are of two types:
 - Organ specific:** are those where antibodies and invasive destructive lesions are directed against one organ of the body
Target organs are- Thyroid, adrenals, stomach and pancreas
Example: Hashimoto's Thyroiditis, Thyrotoxicosis, Myasthenia Gravis
 - Nonorgan specific or Systemic:** are those where antibodies and lesions are directed to antigens which are widespread throughout the body.
Example: Rheumatological disorders characteristically involve the skin, kidney, joints and muscle



what is autoimmunity and classi



All

News

Images

Videos

Books

Sea

Autoimmune diseases can be classified as **systemic or organ-specific** depending on the extent of their clinicopathology (Table 19-2). The systemic category includes systemic lupus erythematosus (SLE), rheumatoid arthritis (RA), scleroderma, primary Sjögren's syndrome, dermatomyositis, and systemic vasculitides.

E [https://www.sciencedirect.com](https://www.sciencedirect.com/topics) › topics

Autoimmunity - an overview | ScienceDirect Topics

About featured snippets

Feedback

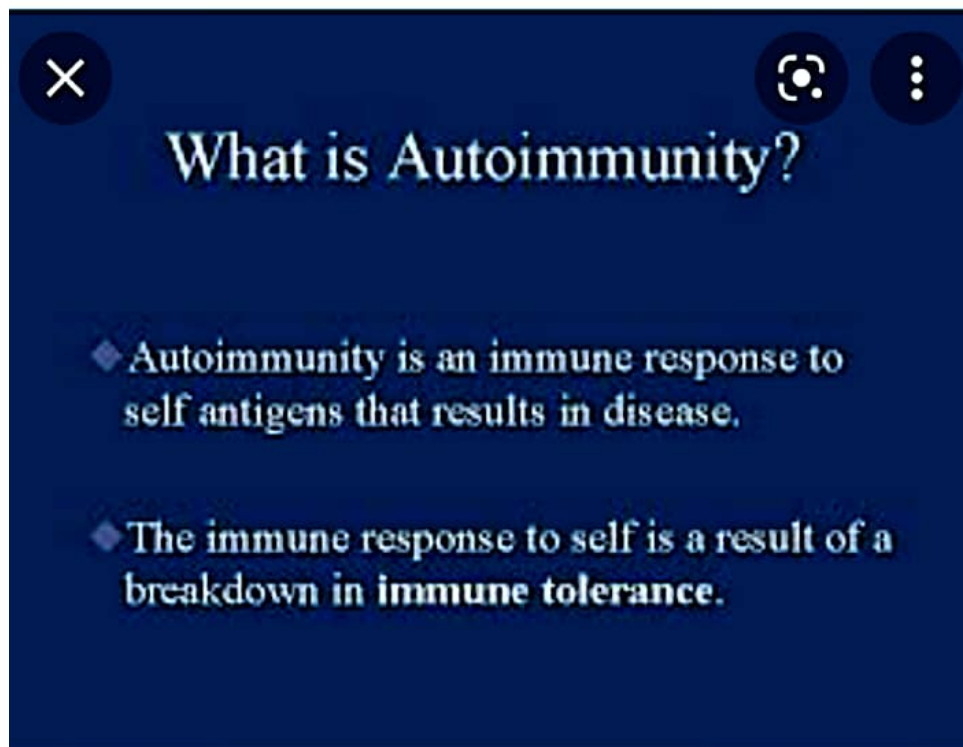


[https://pathology.jhu.edu](https://pathology.jhu.edu/classifica...) › classifica...

Classification of Autoimmune Diseases

Typical systemic autoimmune diseases are





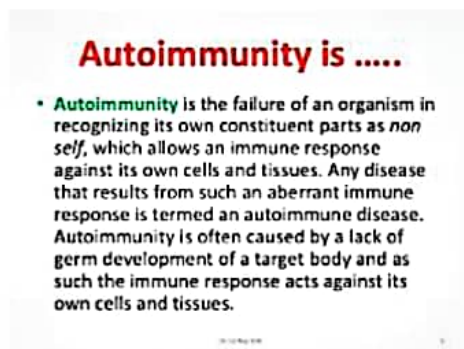
SlideToDoc.com

[Visit](#)

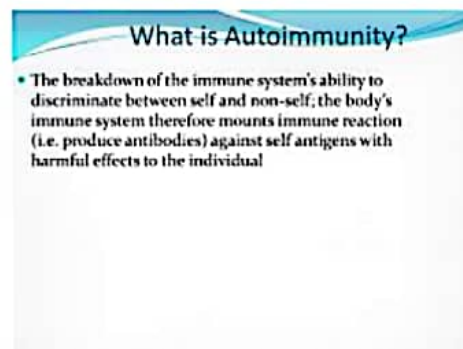
Autoimmunity Robert Beatty
MCB 150 What is Autoimmunity

Images may be subject to copyright. [Learn More](#)

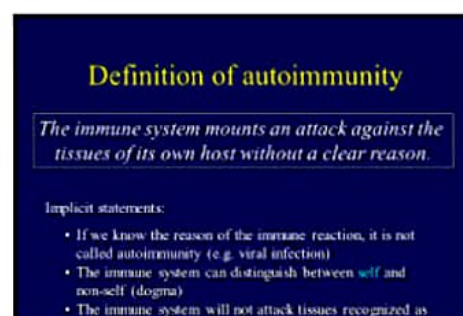
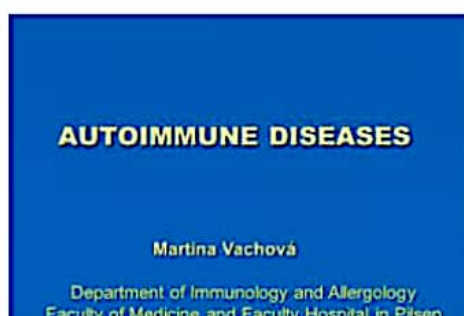
Related images



Autoimmunity
 slideshare.net



Chapter 28 Autoimmune Diso...
 slideplayer.com



Father



Basics



People also ask

What does autoimmunity mean?



What are the examples of autoimmunity?



Common autoimmune disorders include:

- Addison disease.
- Celiac disease - sprue (gluten-sensitive enteropathy)
- Dermatomyositis.
- Graves disease.
- Hashimoto thyroiditis.
- Multiple sclerosis.
- Myasthenia gravis.
- Pernicious anemia.

[More items...](#) • 08-Apr-2019

 <https://medlineplus.gov> › article

[Autoimmune disorders:](#)
[MedlinePlus Medical](#)
[Encyclopedia](#)





what is autoimmunity ?



All

Images

Videos

News

Books

See

Autoimmunity



Sometimes **the immune system** makes a mistake and attacks the body's own tissues or organs. This is called autoimmunity. One example of an autoimmune disease is type 1 diabetes, in which the immune system destroys the cells in the pancreas that produce insulin.

<https://kidshealth.org> › parents › aut...

Definition: Autoimmunity (for Parents) - Nemours Kidshealth

Antibody test



Father



Basics



People also ask





Immune mechanism of rejection



All

Images

Videos

News

Books

See

Showing results for **Immune mechanism of rejection?**

Search instead for **Immune mechanism of rejection?**

Two major immunological mechanisms occur during allograft rejection: the **nonspecific innate response** that predominates in the early phase of the immune response, and the donor-specific adaptive response that results from alloantigen recognition by host T cells.

<https://www.ncbi.nlm.nih.gov/pmc>

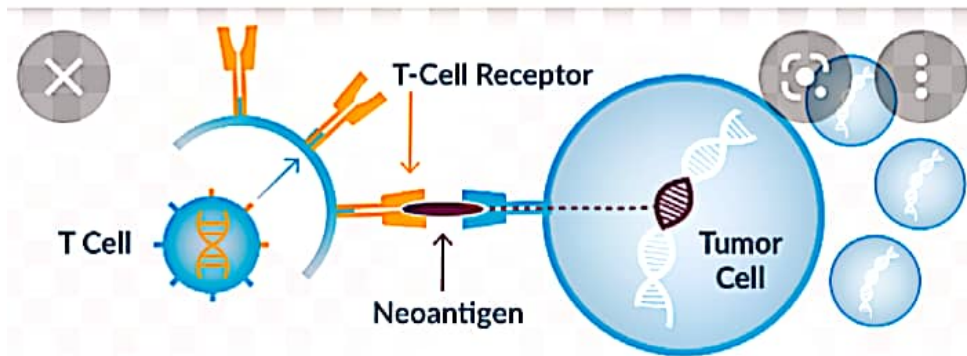
Effector Mechanisms of Rejection - NCBI

About featured snippets

Feedback

People also ask





T-Cell receptors

T-cell receptors or TCRs are molecules on the surface of cancer fighting T cells that have the ability to interrogate individual cancer cells and see beneath the cell membrane.

Neoantigens

Antigens are the unique molecules or proteins that help immune cells identify and fight cancer cells. Neoantigens are unique to each patient's tumor cells.

TCRs have the potential to be genetically modified so T cells can identify the neoantigen signature unique to one patient's cancer cells and then attack them.



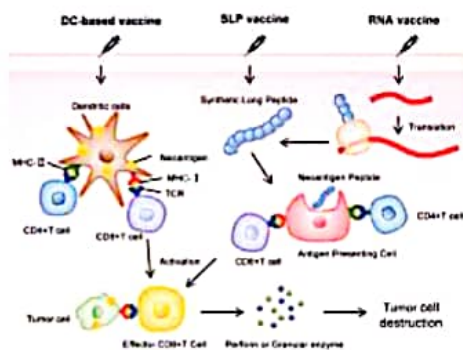
CB Insights

Visit

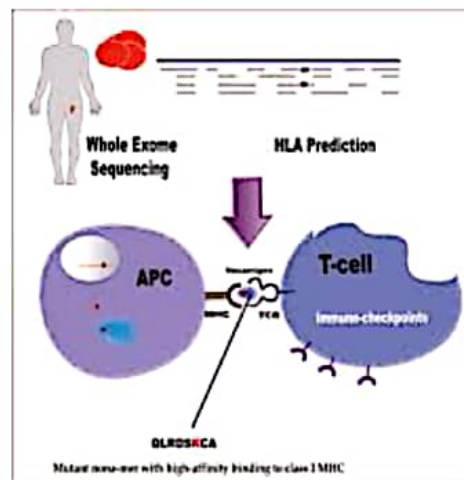
Neoantigen Cancer Vaccines & Companies Bringing it to the...

Images may be subject to copyright. [Learn More](#)

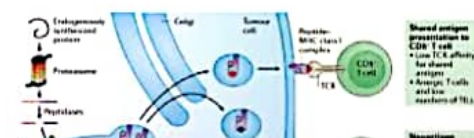
Related images



Neoantigen vaccine: an emer...
molecular-cancer.biomedcentral...



What are Cancer Neoantigens...
weillcornellgucancer.org





use of tumor Neo- antigens?



All

Images

News

Videos

Books

See

Did you mean: [use of tumor **Neoantigens?**](#)

A new protein that forms on cancer cells when certain mutations occur in tumor DNA. Neoantigens may play an important role in helping the body make an immune response against cancer cells. Neoantigens used **in vaccines and other types of immunotherapy** are being studied in the treatment of many types of cancer.

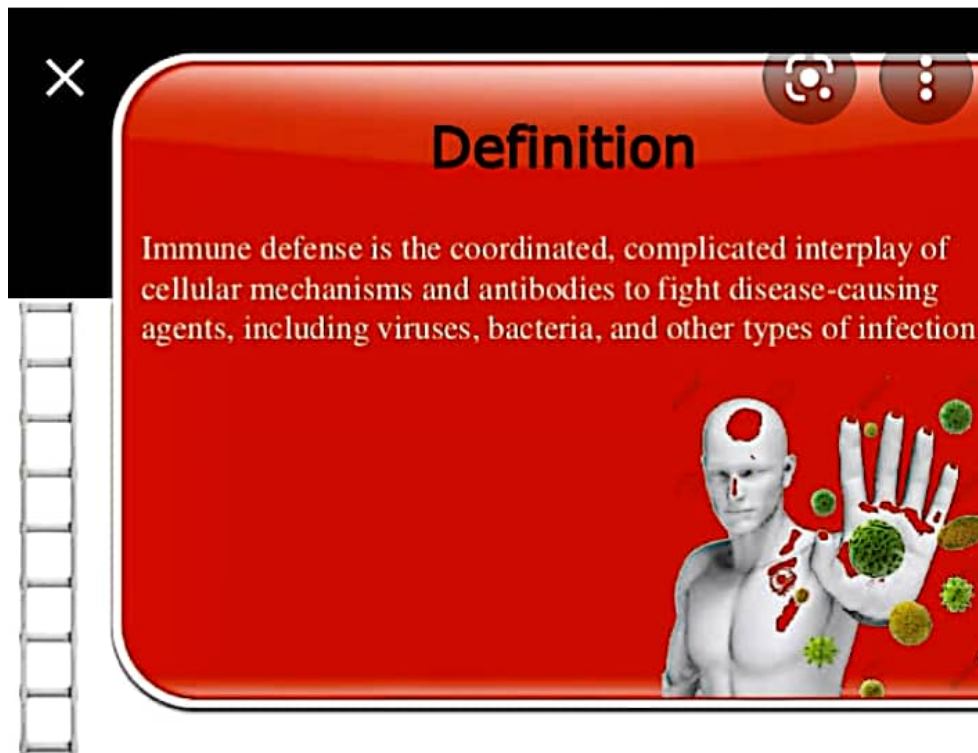
<https://www.cancer.gov/def/nea...>

[Definition of neoantigen - NCI
Dictionary of Cancer Terms](#)

[About featured snippets](#)[Feedback](#)

People also ask





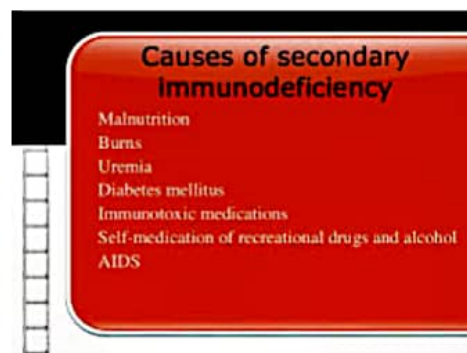
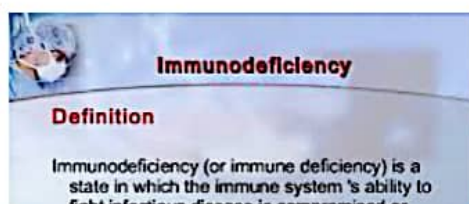
SlideShare

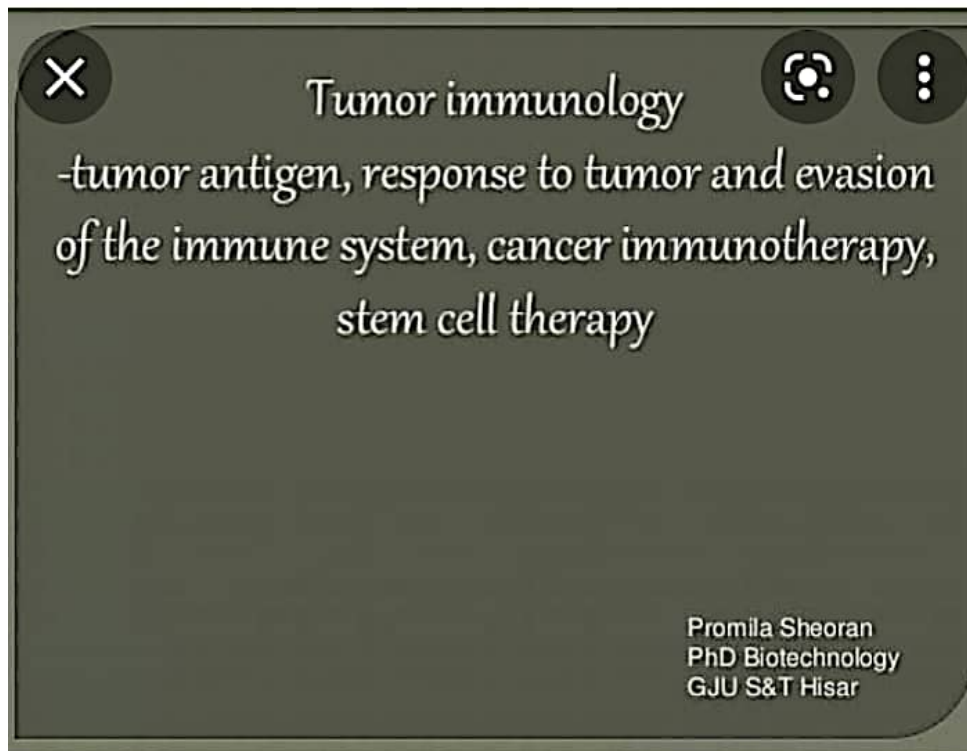
[Visit](#)

immune deficiency

Images may be subject to copyright. [Learn More](#)

Related images

immune deficiency
slideshare.netimmune deficiency
slideshare.net



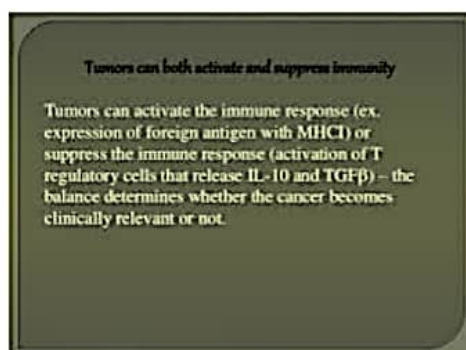
SlideShare

[Visit](#)

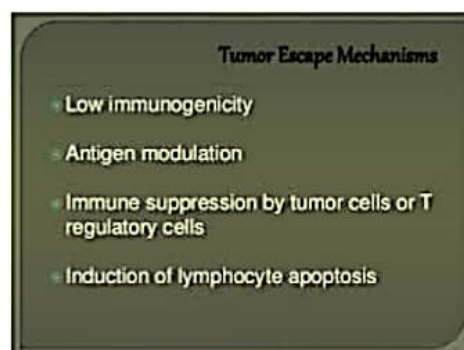
Tumor immunology

Images may be subject to copyright. [Learn More](#)

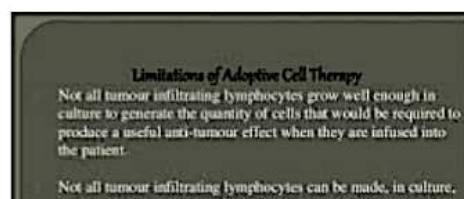
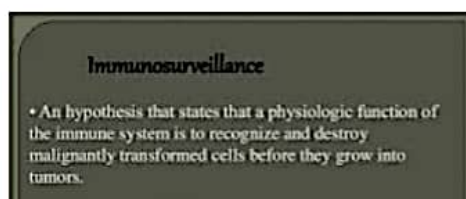
Related images



Tumor immunology
slideshare.net



Tumor immunology
slideshare.net



Tumor Immunology:

Definitions:

-Tumor (neoplasm): unchecked cellular growth.

-Two types:

1-Benign: controlled tumor.

2-Malignant: uncontrolled tumor.

-Malignant tumors are characterized by:

1-**Progressive growth** without exogenous growth signals.

2-**Invasiveness** of normal tissue boundaries.

3-**Metastasis**.

4-Mainly of **Monoclonal** origin.



SlideServe

Visit

PPT - Tumor Immunology:
PowerPoint Presentation, free...

Images may be subject to copyright. [Learn More](#)

Related images

Classification of malignant tumor:

Malignant tumor is classified according to its embryonic origin into different groups:

1-Carcinoma: Malignant tumors develop from endodermal or ectodermal tissues.

Examples: Skin carcinoma, Colon and lung carcinoma.

2-Sarcoma: Malignant tumors develop from bone and cartilage.

3- Leukemia: Malignant tumor of hematopoietic lineage that proliferate as individual cells.

4-Lymphoma: Malignant tumor of hematopoietic lineage that proliferate as solid mass.

TUMOR IMMUNOLOGY

Objectives

- ❖ Know the evidence for immune reactivity to tumor
- ❖ Know the changes in cellular characteristics due to malignancy
- ❖ Know the host components which affect tumor progression
- ❖ Know the tumor cell components which protect it from the immune system
- ❖ Understand the rationale & approaches of tumor immunotherapy

PPT - Tumor Immunology: Po...
slideserve.com

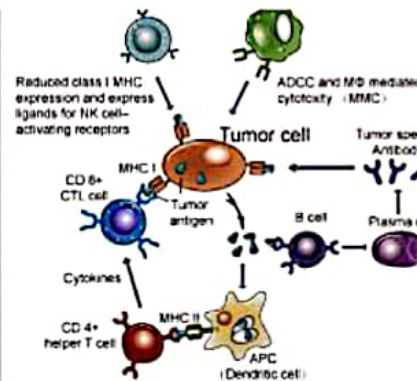
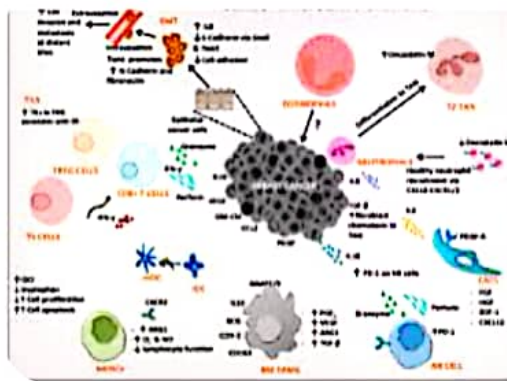
TUMOR IMMUNOLOGY.ppt
studylib.net

Tumor Immunology

- evidence for immune reactivity against tumor
- changes in cellular characteristics due to malignancy
- tumor and host components which affect tumor progression
- use of tumor antigens in diagnosis and immunotherapy

Objectives

- Oncology, prevalence
- Definition, origin, classification, clonal nature
- Tumor growth, vasculatization, necrosis, metastasis, heterogeneity
- Cellular changes, carcinogenesis, oncogens, mutation, tumor supressor genes, viruses
- Tumor immunology, antigens, response to tumor, tumor escape mechanisms
- Immunotherapy, imunodiagnosis,



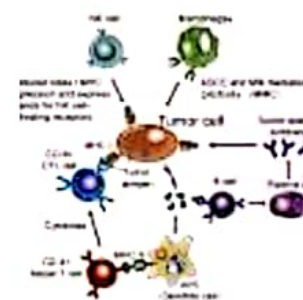
Cancer immunology is an interdisciplinary branch of biology that is concerned with understanding the role of the immune system in the progression and development of cancer; the most well known application is cancer immunotherapy, which utilises the immune system as a treatment for cancer.

[Wikipedia](#)

People also ask

What is meant by tumor immunity?

1. Escaping Immune Recognition by Loss of Antigen Expression. Immune responses to tumor cells impart selective pressures that result in the survival and outgrowth of variant tumor cells with reduced immunogenicity, a process that has been called tumor immunoediting.



[CD https://www.creative-diagnostics.com](https://www.creative-diagnostics.com) > ...

[Tumor Immunity - Creative Diagnostics](#)

with chronic antibody-mediated rejection and graft loss. Moreover, NK cells can prime the adaptive immune system and promote the migration of other immune cells, such as dendritic cells, into the graft leading to an increased allo-immune response and, eventually, to chronic graft rejection. Finally, activated NK cells can infiltrate the transplanted kidney and cause a direct graft damage. Interestingly, immunosuppression can influence NK cell numbers and function, thus causing an increased risk of post-transplant neoplasia or infection. In this review, we will describe how these cells can influence the innate and the adaptive immune response in kidney transplantation and how immunosuppression can modulate NK behavior.

Phenotype, Maturation, Cytotoxic Activity, and Distribution of Natural Killer (NK) Cells



Natural killer (NK) cells are effector





Natural killer cells (NK) represent a population of lymphocytes involved in innate immune response. In addition to their role in anti-viral and anti-tumor defense, they also regulate several aspects of the allo-immune response in kidney transplant recipients. Growing evidence suggests a key role of NK cells in the pathogenesis of immune-mediated graft damage in kidney transplantation. Specific NK cell subsets are associated with operational tolerance in kidney transplant patients. On the other side, allo-reactive NK cells are associated with chronic antibody-mediated rejection and graft loss. Moreover, NK cells can prime the adaptive immune system and promote the migration of other immune cells, such as dendritic cells, into the graft leading to an increased allo-immune response and, eventually, to chronic graft rejection. Finally, activated NK cells can infiltrate the transplanted kidney.





Role of NK cells?

All

Images

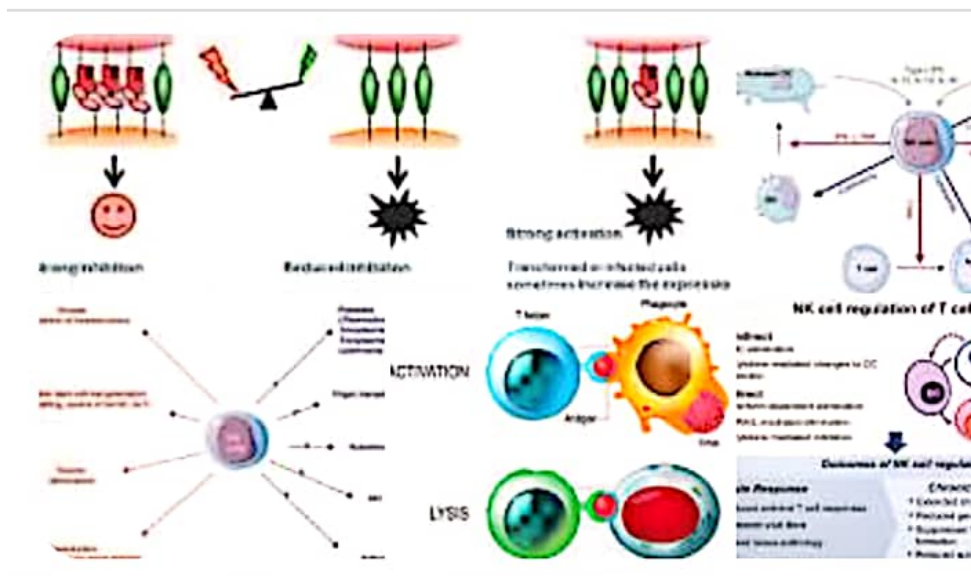
News

Videos

Books

Search

Role of natural killer cells



Natural killer (NK) cells are effector lymphocytes of the innate immune system that **control several types of tumors and microbial infections by limiting their spread and subsequent tissue damage.** 18-Apr-2008


<https://www.nature.com/articles>

Functions of natural killer cells |
Nature Immunology





Images

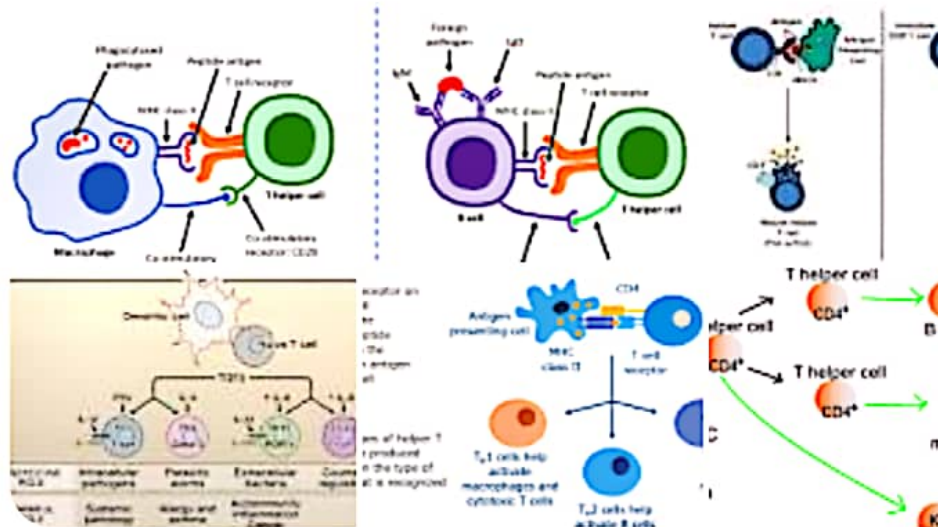
News

Videos

Books

More

Role of helper T cells



Helper T cells are arguably the most important cells in **adaptive immunity**, as they are required for almost all adaptive immune responses. They not only help activate B cells to secrete antibodies and macrophages to destroy ingested microbes, but they also help activate cytotoxic T cells to kill infected target cells.



<https://www.ncbi.nlm.nih.gov/books>

Helper T Cells and Lymphocyte Activation - Molecular - NCBI

About featured snippets

Feedback





important role in adaptive immunity, as they are required for almost all adaptive immune responses. They not only help activate B cells to secrete antibodies and macrophages to destroy ingested microbes, but they also help activate cytotoxic T cells to kill infected target cells.



<https://www.ncbi.nlm.nih.gov/books>

Helper T Cells and Lymphocyte Activation - Molecular - NCBI



About featured snippets



Feedback

People also ask

What is the role of helper T cells in immunity quizlet?



What is the role of helper T cells in the adaptive immune response? Helper T cells activate B cells and **cytotoxic T lymphocytes to kill infected host cells**. Helper T cells directly kill infected host cells. ... Helper T cells also activate cytotoxic T cells, which will search for and destroy infected host cells.



<https://quizlet.com/chapter-16-mastering-questions>

Chapter 16 mastering questions





Search 196,07...



Sign In

...in physiological sciences : an international
journal of physiology produced jointly by the
International Union of Physiological Sciences and
the American Physiological Society

Cytotoxic T lymphocytes mediate lysis of target cells by various mechanisms, including exocytosis of lytic proteins (perforin, granzymes) and receptor-ligand binding of Fas/APO molecules. Death of target cells is characterized by either necrosis or apoptosis, depending on the killing mechanism used and on the metabolism of the target cell itself. [Collapse](#)

[View On Pub...](#)[\[PDF\] Physi...](#)[Save](#)[Alert](#)[Feed](#)

Share This Paper



By clicking accept or continuing to use the site, you agree to the terms outlined in our [Privacy Policy](#), [Terms of Service](#), and [Dataset License](#)

ACCEPT & CONTINUE





Cytotoxic T Lymphocytes (CTL)

Cytotoxic T lymphocytes are [lymphocytes](#) that kill other ("target") cells.

Targets may include:

- virus-infected cells (e.g., [HIV-infected CD4⁺](#) T cells);
- cells infected with intracellular bacterial or protozoal parasites;
- [allografts](#) such as transplanted kidney, heart, lungs, etc.
- cancer cells. (The [tumor-infiltrating lymphocytes](#), **TIL**, that have shown some promise in cancer therapy contain CTLs.)
- In mice, and perhaps in humans, CTLs also can attack CD4⁺ T cells that are autoreactive; that is, T cells that recognize self antigens and thus threaten the host with autoimmunity.
- While the mechanism above protects against autoimmunity, CTLs may trigger some [autoimmune disorders](#), e.g. help destroy the [beta cells](#) of the islets of Langerhans, leading to [Type I diabetes mellitus](#).

Properties of CTLs

Most of them

- belong to the [CD8⁺](#) subset of T cells;
- use the [αβ T-cell receptor for antigen](#) (TCR)
- thus recognize antigens nestled in the

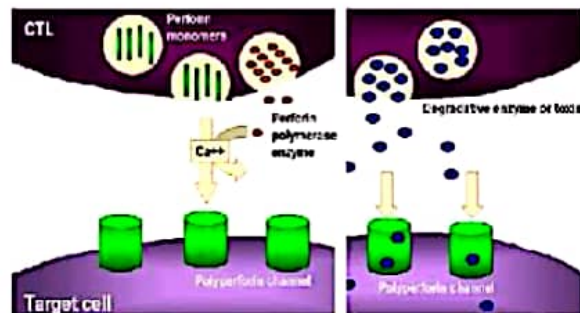




Mechanisms of CTL killing



- CTL granule-mediated killing
 - perforin & granzymes released from CTL granules
 - Perforin polymerizes and forms channels in membrane
 - Granzymes (serine proteases) enter through channel, activate caspases



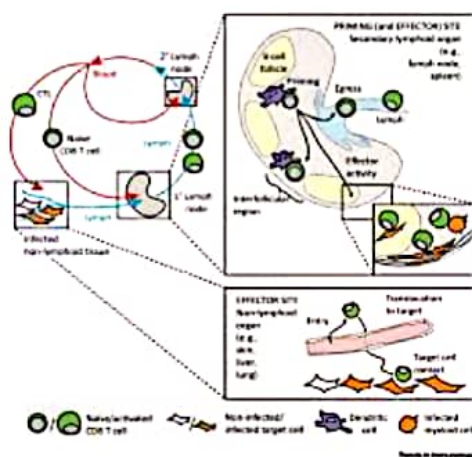
SlideToDoc.com

Visit

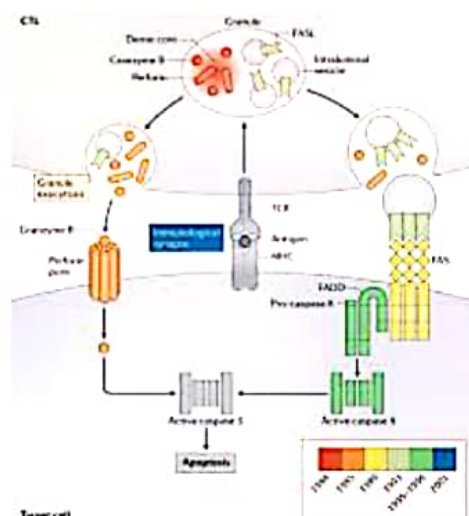
Cellcell interactions in immune responses Teaching objectives To

Images may be subject to copyright. [Learn More](#)

Related images

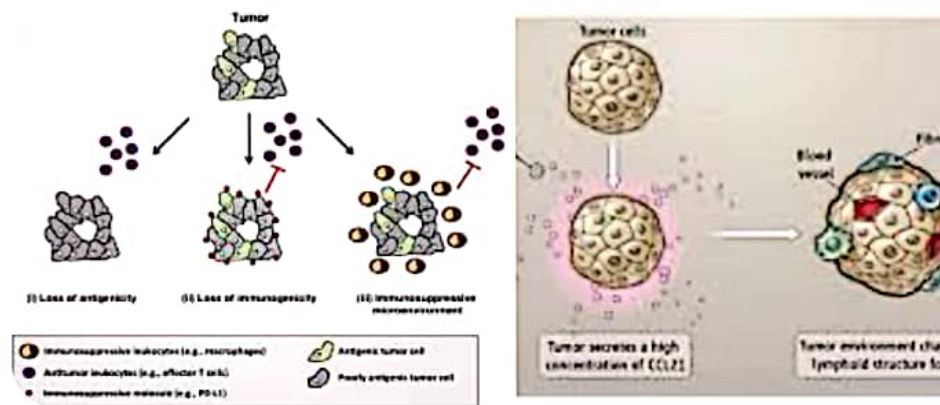


Mechanisms and Dynamics o...
cell.com



An early history of T cell-med...
nature.com





Antigenic escape, immune escape, immune evasion or escape mutation occurs when the immune system of a host, especially of a human being, is unable to respond to an infectious agent, or, in other words, the host's immune system is no longer able to recognize and eliminate a pathogen such as a virus. [Wikipedia](#)

People also ask

What is meaning of immune evasion?



Immune evasion is a **strategy used by pathogenic organisms and tumours to evade a host's immune response to maximize their probability of being transmitted to a fresh host or to continue growing, respectively.**

 <https://www.nature.com> › subjects

[Immune evasion - Latest research and news | Nature](#)

[More results](#)



what is Immune Evasion?

All

Images

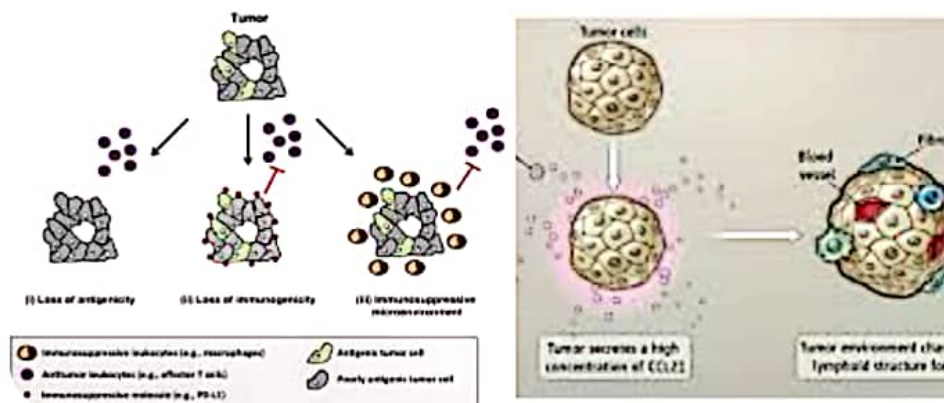
News

Videos

Books

Sea

Antigenic escape



Antigenic escape, immune escape, immune evasion or escape mutation occurs when the immune system of a host, especially of a human being, is unable to respond to an infectious agent, or, in other words, the host's immune system is no longer able to recognize and eliminate a pathogen such as a virus. [Wikipedia](#)

People also ask

What is meaning of immune evasion?





Abstract: Breast cancer (BC) is the most common malignant tumor among women, with high morbidity and mortality. Its onset, development, metastasis, and prognosis vary among individuals due to the interactions between tumors and host immunity. Many diverse mechanisms have been associated with BC, with immune evasion being the most widely studied to date. Tumor cells can escape from the body's immune response, which targets abnormal components and foreign bodies, using different approaches including modification of surface antigens and modulation of the surrounding environment. In this review, we summarize the mechanisms and factors that impact the immunoediting process and analyze their functions in detail.

Keywords: immune evasion, mechanism, breast

In order to provide our website visitors and registered users with a service tailored to their individual preferences we use cookies to analyse visitor traffic and personalise content. You can learn about our use of cookies by reading our [Privacy Policy](#). We also retain data in relation to our visitors and registered users for internal purposes and for sharing information with our business partners. You can learn about what data of yours we retain, how it is processed, who it is shared with and your right to have your data deleted by reading our [Privacy Policy](#).

If you agree to our use of cookies and the contents of our Privacy Policy please click 'accept'.

Accept

Three phases of immune response

Three main phases encompass the immune response that is orchestrated by antigen-specific T cells: **expansion, contraction and memory** (see Fig. 1a).

 <https://www.ncbi.nlm.nih.gov/pmc>

[What are the three phases of immune response? - NCBI](#)

 three phases of immune response

 About featured snippets

 Feedback

People also ask

What are the steps of immune response?



What is meant by Immunoregulation?



A basic property of the immune system is its ability to respond to virtually any foreign antigen but not to react against self components. ... The term 'immunoregulation' usually refers to **such control mechanisms which influence the immune response in a positive or negative way.**

 <https://link.springer.com/pdf>





Immunoregulation following thre

All

Images

News

Videos

Books

Sea

Three phases of immune response

Three main phases encompass the immune response that is orchestrated by antigen-specific T cells: **expansion, contraction and memory** (see Fig. 1a).

 <https://www.ncbi.nlm.nih.gov/pmc>

[What are the three phases of immune response? - NCBI](#)

 three phases of immune response

 About featured snippets

 Feedback

People also ask

What are the steps of immune response?



What is meant by Immunoregulation?



Categories of cytokines

TABLE 4-2 Six Cytokine Families

Family name	Representative members of family	Comments
Interleukin 1 family	IL-1 α , IL-1 β , IL-1Ra, IL-18, IL-33	IL-1 was the first noninterferon cytokine to be identified. Members of this family include important inflammatory mediators.
Hematopoietin (Class I cytokine) family	IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-12, IL-13, IL-15, IL-21, IL-23, GM-CSF, G-CSF, Growth hormone, Prolactin, Erythropoietin/hematopoietin	This large family of small cytokine molecules exhibits striking sequence and functional diversity.
Interferon (Class II cytokine) family	IFN- α , IFN- β , IFN- γ , IL-10, IL-19, IL-20, IL-22, IL-24	While the IFNs have important roles in anti-viral responses, all are important modulators of immune responses.
Tumor Necrosis Factor family	TNF- α , TNF- β , CD40L, Fas (CD95), BAFF, APRIL, LT β	Members of this family may be either soluble or membrane bound; they are involved in immune system development, effector functions, and homeostasis.
Interleukin 17 family	IL-17 (IL17-A), IL17B, C, D, and F	This is the most recently discovered family; members function to promote neutrophil accumulation and activation, and are proinflammatory.
Chemokines (see Appendix III)	IL-8, CCL19, CCL21, RANTES, CCL2 (MCP-1), CCL3 (MIP-1 α)	All serve chemoattractant function.



SlidePlayer

Visit

Cytokines. - ppt video online download

Images may be subject to copyright. [Learn More](#)

Related images

Family member	Other cytokine families	Receptors	Expressed by which cells	Main functions
IL-1 α	IL-1 β and IL-18	IL-1RI and IL-1RAcP	T, T _H 1, T _H 2, T _H 17, T _H 18, T _H 19, T _H 20, T _H 21, T _H 22, T _H 23, T _H 24, T _H 25, T _H 26, T _H 27, T _H 28, T _H 29, T _H 30, T _H 31, T _H 32, T _H 33, T _H 34, T _H 35, T _H 36, T _H 37, T _H 38, T _H 39, T _H 40, T _H 41, T _H 42, T _H 43, T _H 44, T _H 45, T _H 46, T _H 47, T _H 48, T _H 49, T _H 50, T _H 51, T _H 52, T _H 53, T _H 54, T _H 55, T _H 56, T _H 57, T _H 58, T _H 59, T _H 60, T _H 61, T _H 62, T _H 63, T _H 64, T _H 65, T _H 66, T _H 67, T _H 68, T _H 69, T _H 70, T _H 71, T _H 72, T _H 73, T _H 74, T _H 75, T _H 76, T _H 77, T _H 78, T _H 79, T _H 80, T _H 81, T _H 82, T _H 83, T _H 84, T _H 85, T _H 86, T _H 87, T _H 88, T _H 89, T _H 90, T _H 91, T _H 92, T _H 93, T _H 94, T _H 95, T _H 96, T _H 97, T _H 98, T _H 99, T _H 100, T _H 101, T _H 102, T _H 103, T _H 104, T _H 105, T _H 106, T _H 107, T _H 108, T _H 109, T _H 110, T _H 111, T _H 112, T _H 113, T _H 114, T _H 115, T _H 116, T _H 117, T _H 118, T _H 119, T _H 120, T _H 121, T _H 122, T _H 123, T _H 124, T _H 125, T _H 126, T _H 127, T _H 128, T _H 129, T _H 130, T _H 131, T _H 132, T _H 133, T _H 134, T _H 135, T _H 136, T _H 137, T _H 138, T _H 139, T _H 140, T _H 141, T _H 142, T _H 143, T _H 144, T _H 145, T _H 146, T _H 147, T _H 148, T _H 149, T _H 150, T _H 151, T _H 152, T _H 153, T _H 154, T _H 155, T _H 156, T _H 157, T _H 158, T _H 159, T _H 160, T _H 161, T _H 162, T _H 163, T _H 164, T _H 165, T _H 166, T _H 167, T _H 168, T _H 169, T _H 170, T _H 171, T _H 172, T _H 173, T _H 174, T _H 175, T _H 176, T _H 177, T _H 178, T _H 179, T _H 180, T _H 181, T _H 182, T _H 183, T _H 184, T _H 185, T _H 186, T _H 187, T _H 188, T _H 189, T _H 190, T _H 191, T _H 192, T _H 193, T _H 194, T _H 195, T _H 196, T _H 197, T _H 198, T _H 199, T _H 200, T _H 201, T _H 202, T _H 203, T _H 204, T _H 205, T _H 206, T _H 207, T _H 208, T _H 209, T _H 210, T _H 211, T _H 212, T _H 213, T _H 214, T _H 215, T _H 216, T _H 217, T _H 218, T _H 219, T _H 220, T _H 221, T _H 222, T _H 223, T _H 224, T _H 225, T _H 226, T _H 227, T _H 228, T _H 229, T _H 230, T _H 231, T _H 232, T _H 233, T _H 234, T _H 235, T _H 236, T _H 237, T _H 238, T _H 239, T _H 240, T _H 241, T _H 242, T _H 243, T _H 244, T _H 245, T _H 246, T _H 247, T _H 248, T _H 249, T _H 250, T _H 251, T _H 252, T _H 253, T _H 254, T _H 255, T _H 256, T _H 257, T _H 258, T _H 259, T _H 260, T _H 261, T _H 262, T _H 263, T _H 264, T _H 265, T _H 266, T _H 267, T _H 268, T _H 269, T _H 270, T _H 271, T _H 272, T _H 273, T _H 274, T _H 275, T _H 276, T _H 277, T _H 278, T _H 279, T _H 280, T _H 281, T _H 282, T _H 283, T _H 284, T _H 285, T _H 286, T _H 287, T _H 288, T _H 289, T _H 290, T _H 291, T _H 292, T _H 293, T _H 294, T _H 295, T _H 296, T _H 297, T _H 298, T _H 299, T _H 300, T _H 301, T _H 302, T _H 303, T _H 304, T _H 305, T _H 306, T _H 307, T _H 308, T _H 309, T _H 310, T _H 311, T _H 312, T _H 313, T _H 314, T _H 315, T _H 316, T _H 317, T _H 318, T _H 319, T _H 320, T _H 321, T _H 322, T _H 323, T _H 324, T _H 325, T _H 326, T _H 327, T _H 328, T _H 329, T _H 330, T _H 331, T _H 332, T _H 333, T _H 334, T _H 335, T _H 336, T _H 337, T _H 338, T _H 339, T _H 340, T _H 341, T _H 342, T _H 343, T _H 344, T _H 345, T _H 346, T _H 347, T _H 348, T _H 349, T _H 350, T _H 351, T _H 352, T _H 353, T _H 354, T _H 355, T _H 356, T _H 357, T _H 358, T _H 359, T _H 360, T _H 361, T _H 362, T _H 363, T _H 364, T _H 365, T _H 366, T _H 367, T _H 368, T _H 369, T _H 370, T _H 371, T _H 372, T _H 373, T _H 374, T _H 375, T _H 376, T _H 377, T _H 378, T _H 379, T _H 380, T _H 381, T _H 382, T _H 383, T _H 384, T _H 385, T _H 386, T _H 387, T _H 388, T _H 389, T _H 390, T _H 391, T _H 392, T _H 393, T _H 394, T _H 395, T _H 396, T _H 397, T _H 398, T _H 399, T _H 400, T _H 401, T _H 402, T _H 403, T _H 404, T _H 405, T _H 406, T _H 407, T _H 408, T _H 409, T _H 410, T _H 411, T _H 412, T _H 413, T _H 414, T _H 415, T _H 416, T _H 417, T _H 418, T _H 419, T _H 420, T _H 421, T _H 422, T _H 423, T _H 424, T _H 425, T _H 426, T _H 427, T _H 428, T _H 429, T _H 430, T _H 431, T _H 432, T _H 433, T _H 434, T _H 435, T _H 436, T _H 437, T _H 438, T _H 439, T _H 440, T _H 441, T _H 442, T _H 443, T _H 444, T _H 445, T _H 446, T _H 447, T _H 448, T _H 449, T _H 450, T _H 451, T _H 452, T _H 453, T _H 454, T _H 455, T _H 456, T _H 457, T _H 458, T _H 459, T _H 460, T _H 461, T _H 462, T _H 463, T _H 464, T _H 465, T _H 466, T _H 467, T _H 468, T _H 469, T _H 470, T _H 471, T _H 472, T _H 473, T _H 474, T _H 475, T _H 476, T _H 477, T _H 478, T _H 479, T _H 480, T _H 481, T _H 482, T _H 483, T _H 484, T _H 485, T _H 486, T _H 487, T _H 488, T _H 489, T _H 490, T _H 491, T _H 492, T _H 493, T _H 494, T _H 495, T _H 496, T _H 497, T _H 498, T _H 499, T _H 500, T _H 501, T _H 502, T _H 503, T _H 504, T _H 505, T _H 506, T _H 507, T _H 508, T _H 509, T _H 510, T _H 511, T _H 512, T _H 513, T _H 514, T _H 515, T _H 516, T _H 517, T _H 518, T _H 519, T _H 520, T _H 521, T _H 522, T _H 523, T _H 524, T _H 525, T _H 526, T _H 527, T _H 528, T _H 529, T _H 530, T _H 531, T _H 532, T _H 533, T _H 534, T _H 535, T _H 536, T _H 537, T _H 538, T _H 539, T _H 540, T _H 541, T _H 542, T _H 543, T _H 544, T _H 545, T _H 546, T _H 547, T _H 548, T _H 549, T _H 550, T _H 551, T _H 552, T _H 553, T _H 554, T _H 555, T _H 556, T _H 557, T _H 558, T _H 559, T _H 560, T _H 561, T _H 562, T _H 563, T _H 564, T _H 565, T _H 566, T _H 567, T _H 568, T _H 569, T _H 570, T _H 571, T _H 572, T _H 573, T _H 574, T _H 575, T _H 576, T _H 577, T _H 578, T _H 579, T _H 580, T _H 581, T _H 582, T _H 583, T _H 584, T _H 585, T _H 586, T _H 587, T _H 588, T _H 589, T _H 590, T _H 591, T _H 592, T _H 593, T _H 594, T _H 595, T _H 596, T _H 597, T _H 598, T _H 599, T _H 600, T _H 601, T _H 602, T _H 603, T _H 604, T _H 605, T _H 606, T _H 607, T _H 608, T _H 609, T _H 610, T _H 611, T _H 612, T _H 613, T _H 614, T _H 615, T _H 616, T _H 617, T _H 618, T _H 619, T _H 620, T _H 621, T _H 622, T _H 623, T _H 624, T _H 625, T _H 626, T _H 627, T _H 628, T _H 629, T _H 630, T _H 631, T _H 632, T _H 633, T _H 634, T _H 635, T _H 636, T _H 637, T _H 638, T _H 639, T _H 640, T _H 641, T _H 642, T _H 643, T _H 644, T _H 645, T _H 646, T _H 647, T _H 648, T _H 649, T _H 650, T _H 651, T _H 652, T _H 653, T _H 654, T _H 655, T _H 656, T _H 657, T _H 658, T _H 659, T _H 660, T _H 661, T _H 662, T _H 663, T _H 664, T _H 665, T _H 666, T _H 667, T _H 668, T _H 669, T _H 670, T _H 671, T _H 672, T _H 673, T _H 674, T _H 675, T _H 676, T _H 677, T _H 678, T _H 679, T _H 680, T _H 681, T _H 682, T _H 683, T _H 684, T _H 685, T _H 686, T _H 687, T _H 688, T _H 689, T _H 690, T _H 691, T _H 692, T _H 693, T _H 694, T _H 695, T _H 696, T _H 697, T _H 698, T _H 699, T _H 700, T _H 701, T _H 702, T _H 703, T _H 704, T _H 705, T _H 706, T _H 707, T _H 708, T _H 709, T _H 710, T _H 711, T _H 712, T _H 713, T _H 714, T _H 715, T _H 716, T _H 717, T _H 718, T _H 719, T _H 720, T _H 721, T _H 722, T _H 723, T _H 724, T _H 725, T _H 726, T _H 727, T _H 728, T _H 729, T _H 730, T _H 731, T _H 732, T _H 733, T _H 734, T _H 735, T _H 736, T _H 737, T _H 738, T _H 739, T _H 740, T _H 741, T _H 742, T _H 743, T _H 744, T _H 745, T _H 746, T _H 747, T _H 748, T _H 749, T _H 750, T _H 751, T _H 752, T _H 753, T _H 754, T _H 755, T _H 756, T _H 757, T _H 758, T _H 759, T _H 760, T _H 761, T _H 762, T _H 763, T _H 764, T _H 765, T _H 766, T _H 767, T _H 768, T _H 769, T _H 770, T _H 771, T _H 772, T _H 773, T _H 774, T _H 775, T _H 776, T _H 777, T _H 778, T _H 779, T _H 780, T _H 781, T _H 782, T _H 783, T _H 784, T _H 785, T _H 786, T _H 787, T _H 788, T _H 789, T _H 790, T _H 791, T _H 792, T _H 793, T _H 794, T _H 795, T _H 796, T _H 797, T _H 798, T _H 799, T _H 800, T _H 801, T _H 802, T _H 803, T _H 804, T _H 805, T _H 806, T _H 807, T _H 808, T _H 809, T _H 810, T _H 811, T _H 812, T _H 813, T _H 814, T _H 815, T _H 816, T _H 817, T _H 818, T _H 819, T _H 820, T _H 821, T _H 822, T _H 823, T _H 824, T _H 825, T _H 826, T _H 827, T _H 828, T _H 829, T _H 830, T _H 831, T _H 832, T _H 833, T _H 834, T _H 835, T _H 836, T _H 837, T _H 838, T _H 839, T _H 840, T _H 841, T _H 842, T _H 843, T _H 844, T _H 845, T _H 846, T _H 847, T _H 848, T _H 849, T _H 850, T _H 851, T _H 852, T _H 853, T _H 854, T _H 855, T _H 856, T _H 857, T _H 858, T _H 859, T _H 860, T _H 861, T _H 862, T _H 863, T _H 864, T _H 865, T _H 866, T _H 867, T _H 868, T _H 869, T _H 870, T _H 871, T _H 872, T _H 873, T _H 874, T _H 875, T _H 876, T _H 877, T _H 878, T _H 879, T _H 880, T _H 881, T _H 882, T _H 883, T _H 884, T _H 885, T _H 886, T _H 887, T _H 888, T _H 889, T _H 890, T _H 891, T _H 892, T _H 893, T _H 894, T _H 895, T _H 896, T _H 897, T _H 898, T _H 899, T _H 900, T _H 901, T _H 902, T _H 903, T _H 904, T _H 905, T _H 906, T _H 907, T _H 908, T _H 909, T _H 910, T _H 911, T _H 912, T _H 913, T _H 914, T _H 915, T _H 916, T _H 917, T _H 918, T _H 919, T _H 920, T _H 921, T _H 922, T _H 923, T _H 924, T _H 925, T _H 926, T _H 927, T _H 928, T _H 929, T _H 930, T _H 931, T _H 932, T _H 933, T _H 934, T _H 935, T _H 936, T _H 937, T _H 938, T _H 939, T _H 940, T _H 941, T _H 942, T _H 943, T _H 944, T _H 945, T _H 946, T _H 947, T _H 948, T _H 949, T _H 950, T _H 951, T _H 952, T _H 953, T _H 954, T _H 955, T _H 956, T _H 957, T _H 958, T _H 959, T _H 960, T _H 961, T _H 962, T _H 963, T _H 964, T _H 965, T _H 966, T _H 967, T _H 968, T _H 969, T _H 970, T _H 971, T _H 972, T _H 973, T _H 974, T _H 975, T _H 976, T _H 977, T _H 978, T _H 979, T _H 980, T _H 981, T _H 982, T _H 983, T _H 984, T _H 985, T _H 986, T _H 987, T _H 988, T _H 989, T _H 990, T _H 991, T _H 992, T _H 993, T _H 994, T _H 995, T _H 996, T _H 997, T _H 998, T _H 999, T _H 1000, T _H 1001, T _H 1002, T _H 1003, T _H 1004, T _H 1005, T _H 1006, T _H 1007, T _H 1008, T _H 1009, T _H 1010, T _H 1011, T _H 1012, T _H 1013, T _H 1014, T _H 1015, T _H 1016, T _H 1017, T _H 1018, T _H 1019, T _H 1020, T _H 1021, T _H 1022, T _H 1023, T _H 1024, T _H 1025, T _H 1026, T _H 1027, T _H 1028, T _H 1029, T _H 1030, T _H 1031, T _H 1032, T _H 1033, T _H 1034, T _H 1035, T _H 1036, T _H 1037, T _H 1038, T _H 1039, T _H 1040, T _H 1041, T _H 1042, T _H 1043, T _H 1044, T _H 1045, T _H 1046, T _H 1047, T _H 1048, T _H 1049, T _H 1050, T _H 1051, T _H 1052, T _H 1053, T _H 1054, T _H 1055, T _H 1056, T _H 1057, T _H 1058, T _H 1059, T _H 1060, T _H 1061, T _H 1062, T _H 1063, T _H 1064, T _H 1065, T _H 1066, T _H 1067, T _H 1068, T _H 1069, T _H 1070, T _H 1071, T _H 1072, T _H 1073, T _H 1074, T _H 1075, T _H 1076, T _H 1077, T _H 1078, T _H 1079, T _H 1080, T _H 1081, T _H 1082, T _H 1083, T _H 1084, T _H 1085, T _H 1086, T _H 1087, T _H 1088, T _H 1089, T _H 1090, T _H 1091, T _H 1092, T _H 1093, T _H 1094, T _H 1095, T _H 1096, T _H 1097, T _H 1098, T _H 1099, T _H 1100, T _H 1101, T _H 1102, T _H 1103, T _H 1104, T _H 1105, T _H 1106, T _H 1107, T _H 1108, T _H 1109, T _H 1110, T _H 1111, T _H 1112, T _H 1113, T _H 1114, T _H 1115, T _H 1116, T _H 1117, T _H 1118, T _H 1119, T _H 1120, T _H 1121, T _H 1122, T _H 11	



CYTOKINES VERSUS INTERLEUKINS



CYTOKINES	INTERLEUKINS
A number of substances, such as interferon, interleukin, and growth factors, which are secreted by certain cells of the immune system and have an effect on other cells	Any of a class of proteins produced by leucocytes for regulating immune responses
A large group of signaling molecules	A subfamily of cytokines
Can have either autocrine, paracrine or endocrine action	Have either autocrine or paracrine action
Regulate the nature, intensity, and duration of the immune response	Induce the proliferation, differentiation, maturation, migration, and adhesion of cells of the immune system
	Visit www.PEDIAA.com

 Pediaa.Com

[Visit](#)

What Is the Difference Between Cytokines and Interleukins -...

Images may be subject to copyright. [Learn More](#)

Related images





CYTOKINES

Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes



SlideShare

Visit

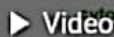
Lecture cytokines

Images may be subject to copyright. [Learn More](#)

Related images

Cytokines

1. Introduction of cytokines
2. Classification of cytokines
3. Cytokine receptors
4. Biological activities of cytokines
5. Biological products derived from cytokines and their receptors



Video

What are Cytokines?

- A collection of polypeptides used for communications between cells
- Play role similar to hormones (messengers of the endocrine system)
 - Hormones usually act at a distance
 - Cytokines act locally
- Differ from growth factors that are produced constitutively, while cytokine production is carefully regulated
- Play an important role in both innate and adaptive immunity

Cytokines 1. Introduction of c...
slideplayer.com

CYTOKINES. - ppt download
slideplayer.com

Naming of Cytokines

- 1. **Monokines** - produced by mononuclear phagocytes (monocytes)
- 2. **Lymphokines** - produced by activated T cells, primarily helper T cells
- 3. **Interleukins** - cytokines made by one leukocyte and acting on other leukocytes
- 4. **Chemokines**-cytokines with chemotactic activities

TOPICS

- DEFINITION
- PROPERTIES OF CYTOKINES
- NOMENCLATURE
- CLASSIFICATION
- CYTOKINE RECEPTORS
- CYTOKINE SIGNALING
- CYTOKINE ANTAGONIST
- TH1 AND TH2
- CYOKINE RELATED DISEASES
- CYTOKINE THERAPY
- CYTOKINE IN HEMATOPOIESIS



Types Of Cytokines

- **Lymphokines** - Cytokines secreted by lymphocytes.
- **Monokines** - Cytokines secreted by monocytes and macrophages.
- **Interleukins** - Cytokines secreted by some leukocytes and acts upon other leukocytes.
- **Interferons** - Cytokines secreted in response to viral infections or other stimuli.
- **Tumor necrosis factors** - These are secreted by some Macrophages and they induces the necrosis of tumor cells and pro inflammatory actions
- **Chemokines** - Low molecular wt. cytokines that affect chemotaxis and other aspects of leukocytes behaviour.



SlidePlayer

Visit

Cytokines Basic introduction.
Contents Definition General ...

Images may be subject to copyright. [Learn More](#)

Related images

Action of Cytokines

- **Pleiotropy**
 - Affects multiple cell types
- **Redundancy**
 - Multiple cytokines affects cells of the same type
- **Synergy**
 - Cytokines acting in concert on the same cell
- **Antagonism**
 - Competing actions
- **Cascading**
 - Cytokines acting sequentially

Cytokines

1. Introduction of cytokines
2. Classification of cytokines
3. Cytokine receptors
4. Biological activities of cytokines
5. Biological products derived from cytokines and their receptors

▶ Video

Cytokines. - ppt video online ...
slideplayer.com

Cytokines 1. Introduction of c...
slideplayer.com

CYTOKINES

- The term cytokine is a general term used to describe a large group of proteins but there are other terms that are commonly used to

Nomenclature...

- **Interleukins** - that act as mediators between leukocytes. The vast majority of these are produced by T-helper cells
- **Tumour necrosis factors** - secreted by macrophages

slideplayer.com

×

SlidePlayer



Ad · <http://www.quansysbio.com/>



Quality Imaging/Better Results - Multiplex Array Technology

The Multiplex Array Kit Is A Sensitive Tool For
Quantifying Diagnostic Antigens. Select Up To 4...



Call +1 435-752-0531

Dictionary

Definitions from Oxford Languages

Search for a word



cytokine

/ˈsaɪtə(ʊ)klaɪn/

noun **PHYSIOLOGY**

plural noun: **cytokines**

any of a number of substances, such as
interferon, interleukin, and growth factors, which
are secreted by certain cells of the immune
system and have an effect on other cells.

Translate cytokines to

Basque ▼

1. zitokinak



Classification of Hypersensitivity			
	Type	Mechanism	Example
Anaphylactic hypersensitivity	I	IgE mediated	Systemic anaphylaxis eg peanut allergy Asthma
Cytotoxic hypersensitivity	II	Antibody mediated	Haemolytic disease of the newborn
Immune complex hypersensitivity	III	Immune complex mediated	Arthus reaction Glomerular nephritis
Cell-mediated hypersensitivity	IV	Cell mediated (delayed)	Contact dermatitis



SlidePlayer

[Visit](#)

Hypersensitivity Type III and IV. Classification of ...

Images may be subject to copyright. [Learn More](#)

Related images

Circulating immune complexes		
Cause	Antigen	Place of deposition
chronic infection	microbial Ag	site of infection, kidneys
autoimmunity	autoantigen	kidneys, joints, vessels, skin
external	environmental Ag	lungs

Hypersensitivity reactions - p...
slideplayer.com

Hypersensitivity reactions

<http://sema.sote.hu/depts/pathophysiology>

László L. Tornóci
Semmelweis University
Institute of Pathophysiology

▶ Video

Hypersensitivity reactions - p...
slideplayer.com

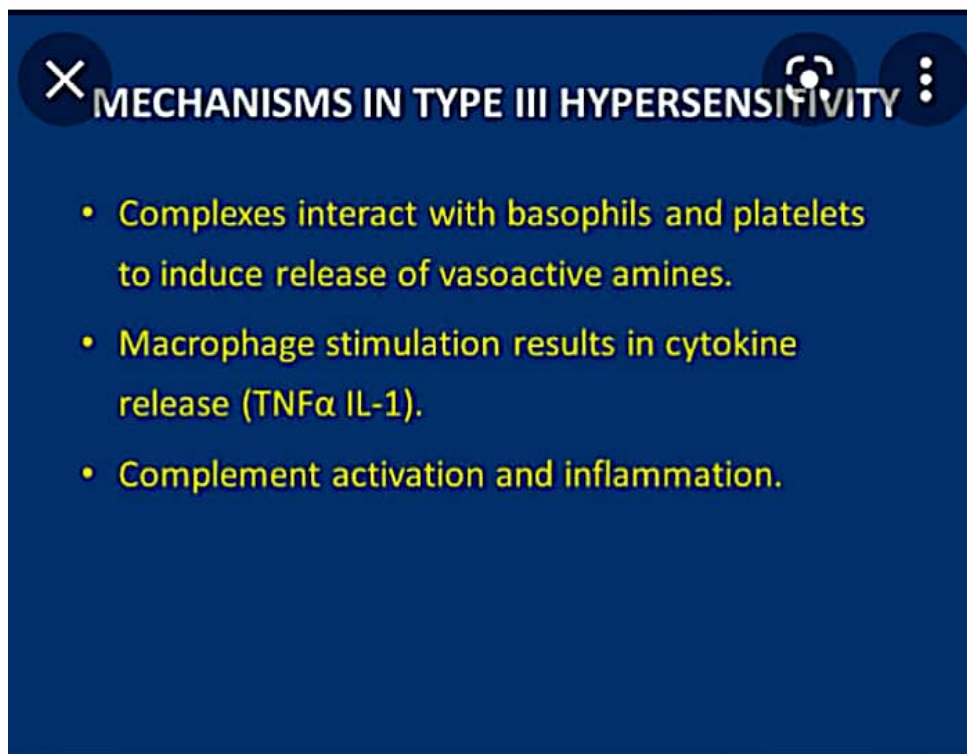
Disorders of Immunity	
A- Mechanisms of immune-mediated injury	

3 Basic Categories			
Cause	1) Persistent	2) Autoimmunity	3) Inhaled

slideplayer.com



SlidePlayer



MECHANISMS IN TYPE III HYPERSENSITIVITY

- Complexes interact with basophils and platelets to induce release of vasoactive amines.
- Macrophage stimulation results in cytokine release (TNF α IL-1).
- Complement activation and inflammation.



SlidePlayer

[Visit](#)

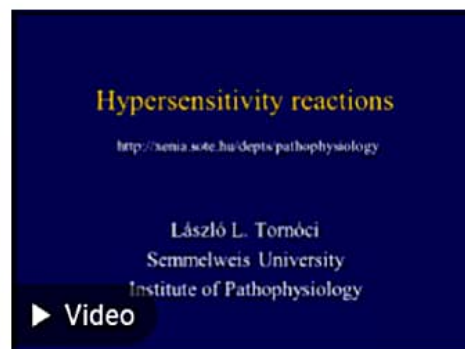
Hypersensitivity Type III and IV. Classification of ...

Images may be subject to copyright. [Learn More](#)

Related images



Immunopathology lecture 2+...
slideshare.net



Hypersensitivity reactions - p...
slideplayer.com



slideplayer.com



SlidePlayer





Hypersensitivity

- **Hypersensitivity reactions** refer to adaptive immune responses that occur in exaggerated or inappropriate forms and result in disease.
- **Four types** of hypersensitivity reactions.



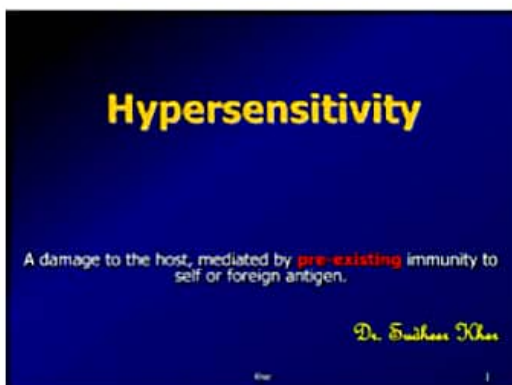
SlidePlayer

Visit

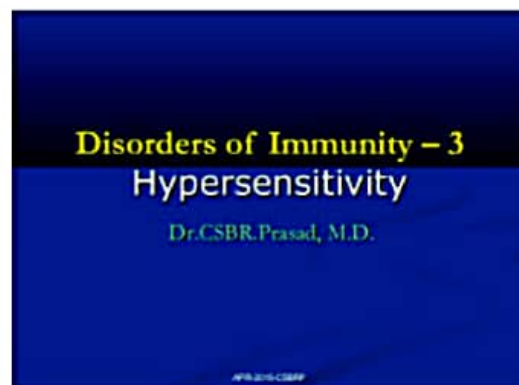
Aims Explain the mechanisms of hypersensitivity reactions. Defi...

Images may be subject to copyright. [Learn More](#)

Related images



PPT - Hypersensitivity Power...
slideserve.com



3 immunology-csbrp
slideshare.net





Types of Hypersensitivity:



Hypersensitivity reactions are divided according to mechanism of action into four groups:

1-Type I (Immediate hypersensitivity).

2-Type II (Cytotoxic hypersensitivity).

3-Type III (Immune complex hypersensitivity).

4-Type IV (Cell-mediated hypersensitivity).



SlidePlayer

Visit

Hypersensitivity Reactions: Definitions: Hypersensitivity ...

Images may be subject to copyright. [Learn More](#)

Related images

Hypersensitivity Reactions:

Definitions:

Hypersensitivity reactions:

inflammatory immune responses induced by repeated antigen exposure resulting in host tissue damage.

Allergen:

is a nonparasitic antigen capable of stimulating hypersensitivity reactions.

Hypersensitivity reactions

- ❖ excessive undesirable (damaging, discomfort producing and sometimes fatal) reactions produced by the normal immune system.
- ❖ require a pre-sensitized (immune) state of the host.
- ❖ Hypersensitivity reactions can be divided into four types:
 1. type I
 2. type II
 3. type III
 4. type IV

Hypersensitivity Reactions: D...
slideplayer.com

Hypersensitivity reactions - p...
slideplayer.com





Translate Hypersensitivity to

Basque ▼

1. Hipersentsibilitatea

[Feedback](#)

[More definitions and word origin ▼](#)

People also ask

What is a hypersensitivity? ^

Hypersensitivity reactions (HR) are **immune responses that are exaggerated or inappropriate against an antigen or allergen**. Coombs and Gell classified hypersensitivity reactions into four forms. 11-Jun-2021

» <https://pubmed.ncbi.nlm.nih.gov> > ...

[Immediate Hypersensitivity Reactions - PubMed](#)

[More results](#)

What is the best definition of hypersensitivity? ▼





What is MASP in Immunology



All

Images

News

Videos

Books

See

Function. MASP-1 is a **serine protease that functions as a component of the lectin pathway of complement activation**. The complement pathway plays an essential role in the innate and adaptive immune response. ... MASP-1 is also able to cleave fibrinogen and factor XIII and may be involved in coagulation.

Aliases: MASP1, 3MC1, CRARF, CRARF1, MAP1, MASP, MASP3, MAp44, PRSS5, ...

External IDs: [OMIM](#): 600521 [HomoloGene](#): 89143 [GeneCards](#): MASP1

Location (UCSC): Chr 3: 187.22 – 187.29 Mb

Species: Human

[W https://en.m.wikipedia.org](https://en.m.wikipedia.org/wiki/MASP1) > wiki

[MASP1 \(protein\) - Wikipedia](https://en.m.wikipedia.org/wiki/MASP1)



About featured snippets



Feedback





The complement system



- A defensive system consisting of over 30 proteins synthesized by hepatocytes, blood monocytes, epithelial cells of the GIT and tissue macrophages . found in circulating blood serum.
- The four major functions of complement include
 1. Opsonization
 2. Inflammation
 3. Target Cytolysis



SlidePlayer

[Visit](#)

The Complement system.

objectives Factors involved in t...

Images may be subject to copyright. [Learn More](#)

Related images

THE COMPLEMENT SYSTEM

- The complement system is a set of plasma proteins that act in a cascade to attack and kill extracellular pathogens.
- Approximately 30 components:
 - activating molecules
 - regulator factors
 - complement receptors
 - membrane proteins which inhibit the lysis of host cells
- Most of the complement proteins and glycoproteins are produced in the [liver](#) in an inactive form (zymogen). Activation is induced by proteolytic cleavage.

General properties

- Present in serum of all animals but its concentration is maximum in serum of guinea pig.
- Complement of one species are able to react with antibodies of other species but not to the same extent.
- C- proteins constitute about 5% of normal serum protein
- Are glycoproteins.
- Are synthesized rapidly in inflammatory responses – hence are called acute phase proteins.
- Heat labile and lost activity at 56°C for 30 mins and inactivated. Immunoglobulins are not inactivated at this temperature.
- Binds with Fc portion of immunoglobulins .

Santosh

THE COMPLEMENT SYSTEM....
slideplayer.com

Complement system
slideshare.net

slideplayer.com



SlidePlayer



Introduction. The complement system helps or “**complements**” the ability of **antibodies and phagocytic cells to clear pathogens from an organism**. It is part of the innate immune system. The complement system consists of a number of small proteins found in the blood, made by the liver.

 <https://www.physio-pedia.com> › Co...

Complement System – Physiopedia

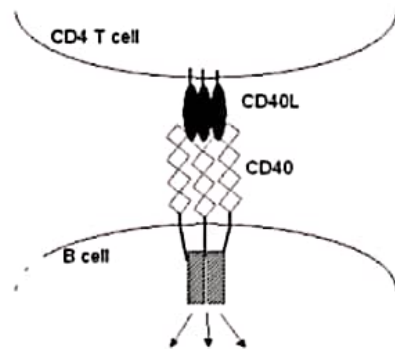


About featured snippets



Feedback

Role of CD40 in B cell activation



Increased expression of cell cycle molecules, survival molecules, promotes isotype switching

- TCR triggering up-regulates CD40L on T cell
- CD40 signaling promotes B cell activation, isotype switching
- CD40 also important in DC, Macrophage function
- CD40L-deficiency = **'hyper-IgM syndrome' (X-linked)**

ICOS – ICOSL also required for Germinal Center responses

ICOS deficiency is a cause of Common Variable Immunodeficiency (CVID)



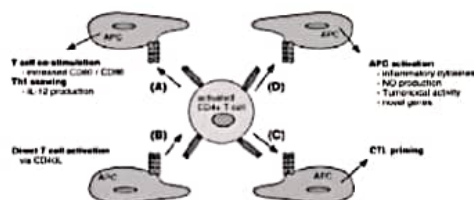
SlidePlayer

Visit

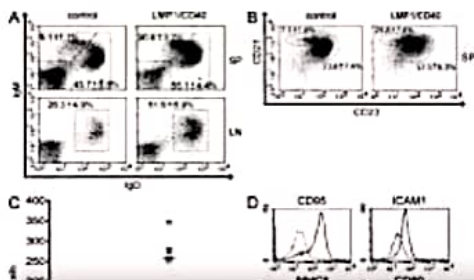
Activation of B cells and Production of Antibodies - ppt...

Images may be subject to copyright. [Learn More](#)

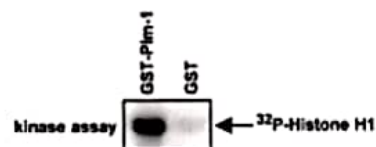
Related images



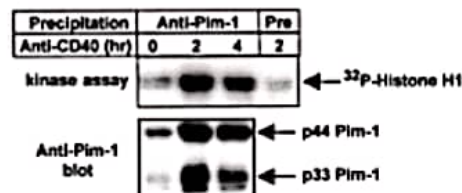
Role of CD40-CD40L in the int...
researchgate.net



A



B



CD40 Signaling in B Cells Reg...
jimmunol.org

slideplayer.com



SlidePlayer



CD40 marker - Guides, protocols and pathways - abcam.com

Wide species reactivity, highly cited and reviewed CD40 antibodies. Specificity guaranteed. View datasheets.

[KO validated antibodies](#)

[Primary Conjugates](#)

[Recombinant antibodies](#)

CD40-activated APCs **up-regulate expression of B7 and MHC class II molecules** (2, 13, 14, 15) and produce inflammatory cytokines, such as IL-12 (16, 17, 18). In turn, these molecules facilitate T cell activation and direct T cell differentiation (19). CD40 is also an important costimulatory molecule for B cells. 01-Dec-2003



<https://www.jimmunol.org> › content

[CD40, but Not CD154, Expression on B Cells Is Necessary for ...](#)





T- dependant	T- independent antigens
- Antigen presenting cells recognize antigen & present it to T-helper cells - T-helper cells stimulate B-cells specific for that antigen to become plasma cells - T- dependant antigens are mainly proteins on viruses, bacteria & other foreign materials.	-Antigens are mainly polysaccharides or lipopolysaccharides with repeating subunits (bacterial capsules). -Immune responses induce the production of IgM of low affinity for the antigen and no immunologic memory
Require T-helper cells	Do not require T-helper cells



KSUMSC

[Visit](#)

Foundation Block Lecture four Antibody mediated immunity

Images may be subject to copyright. [Learn More](#)

Related images

TABLE 11-4 Comparison of primary and secondary antibody responses

Property	Primary response	Secondary response
Responding B cell	Naive (first) B cell	Memory B cell
Lag period following antigen administration	Generally 5-7 days	Generally 1-3 days
Time of peak response	7-10 days	5-7 days
Magnitude of peak antibody response	Naive (depending on antigen)	Generally 100-1000 times higher than primary response
Antibody produced	IgM predominates early in the response	IgG predominates
Antigen	Naive (dependent and duration independent)	Naive (dependent)
Antibody affinity	Lower	Higher

Chapter 11

www2.nau.edu

TABLE 20-2 Types of acquired immunity

Type	Exposure	Antibodies
Artificially acquired active immunity	Exposure to vaccine containing weakened or killed pathogens or their components	Immune response is stimulated without disease (synergist)
Artificially acquired passive immunity	Injection of gamma globulin containing antibodies	Short-term immunity without an immune response
Naturally acquired active immunity	Exposure to live pathogens	Immune response is stimulated with disease symptoms
Naturally acquired passive immunity	Antibodies passed from mother who has active immunity to her fetus or to newborn via breast milk from mother who has active immunity	Short-term antibody for newborn without an immune response

Humoral Immune Response - ... pharmacy180.com

Differences between T cell dependent and T cell independent antigens

T independent Antigens	T dependent Antigen
Structurally simple- LPS, capsular polysaccharide, flagella	Structurally complex- protein in nature
Does dependent immunogenicity	Immunogenic over wide range of dose
No memory	Memory present
No antigen processing	Antigen processing step is needed
Slowly metabolized	Rapidly metabolized

Antigen Learning objectives B... slidetodoc.com

	TD Antigens	TI Antigens
Chemical nature	Proteins	Polymers, antigens, especially polysaccharides, also glycolipids, nucleic acids
Features of Response		
Primary B cell subset	Anticardiac 0 (B2) cells	Anticardiac 0 (B1) cells
Clonal center formation	Yes	No
Secondary isotype (switching)	Yes, IgM, IgG, and IgA	Little, some IgG and IgA
High affinity Ab's (affinity maturation)	Yes	No
Secondary response and memory B cells	Yes	Limited, only for some antigens
Long lasting serum antibody (long lived B cells)	Yes	No/limited

T dependent and t independe... slideshare.net



Differences between T cell dependent and T cell independent antigens

T Independent Antigen	T dependent Antigen
Structurally simple- LPS, capsular polysaccharide, flagella	Structurally complex- protein in nature
Dose dependent Immunogenicity	Immunogenic over wide range of dose
No memory	Memory present
No antigen processing	Antigen processing step is needed
Slowly metabolized	Rapidly metabolized



SlideToDoc.com

Visit

Antigen Learning objectives By
the end of this

Images may be subject to copyright. [Learn More](#)

Related images

Immunogenicity

Ability of an antigen to induce immune response in the body (both humoral and/or cell mediated).

- B cells + antigen → effector B cells (plasma cell) + memory B cells
- T cells + antigen → effector T cells (helper T cell or cytotoxic T cell) + memory T cells

Substance that satisfies this property i.e. immunogenicity - more appropriately called as 'immunogen' rather than 'antigen'.

Antigen Learning objectives B...
slidetodoc.com

T-dependent	T-independent antigens
- Antigen presenting cells recognize antigen & present it to T-helper cells - T-helper cells stimulate B-cells specific for that antigen to become plasma cells - T-dependent antigens are mainly proteins on viruses, bacteria & other foreign materials.	- Antigens are mainly polysaccharides or lipopolysaccharides with repeating subunits (bacterial capsules). - Immune responses induce the production of IgM of low affinity for the antigen and no immunologic memory
Require T-helper cells	Do not require T-helper cells

Foundation Block Lecture fou...
ksumsc.com

TABLE 11-4 Comparison of primary and secondary antibody responses

Property	Primary response	Secondary response
Responding B cell	Naive (virgin) B cell	Memory B cell
Log period following antigen administration	Generally 4-7 days	Generally 1-3 days
Time of peak response	7-10 days	5-6 days
Magnitude of peak antibody response	Varies depending on antigen	Generally 100-1000 times higher than primary response
Isotype produced	IgM predominates early in the response	IgG predominates
Antigens	Thymus-dependent and thymus-independent	Thymus-dependent
Antibody affinity	Lower	Higher

Chapter 11
www2.nau.edu

ANTIGEN AND HOST RELATIONSHIP

Based on the antigen-host relationship, antigens can be grouped into two groups:

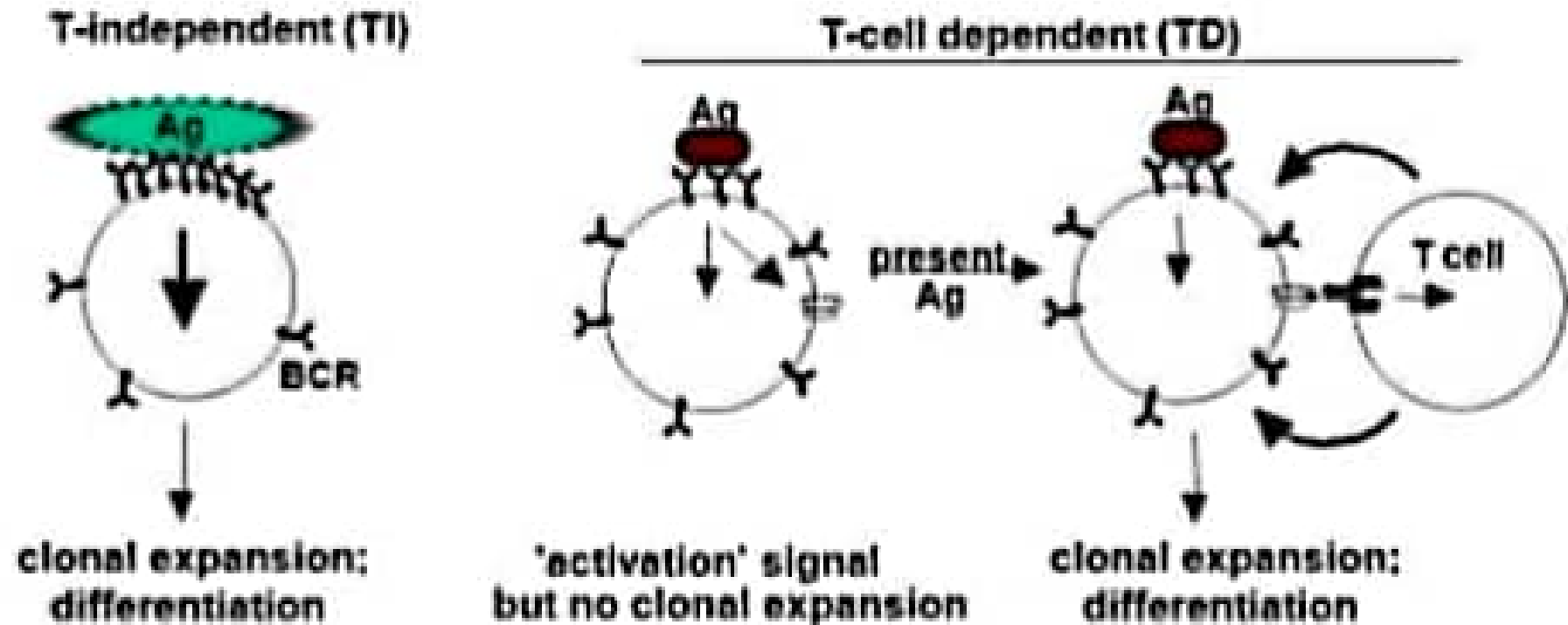
- T-dependent antigens
- T-independent antigens

Antigen Learning objectives B...
slidetodoc.com

T-cell dependent and T-cell independent Antigens:

	T-cell dependent antigen	T-cell independent Antigen
Structural properties	Complex	Simple

Types of B cell Antigens: T-independent and T-dependent



- T-independent antigens are multivalent (e.g. bacterial polysaccharides or repeating determinants on the surface of viruses)
 - fast (within 1-2 days) and predominantly IgM
 - weak in infants and young children
- T-dependent antigens must contain a protein component (true of most antigens) so that T cell help can be received
 - slower (initiate over several days), involve all Ig isotypes (IgM, IgG, IgA, IgE)
 - can lead to antibody affinity maturation and memory



Define Accessory molecules.



All

Images

News

Videos

Books

Sea

Accessory molecules are **cell surface proteins that promote or suppress the response of immune cells**. ... Accessory molecules include cell surface molecules involved in immune cell activation and differentiation, in inhibition of immune cell activity and in cell homing and adhesion.

<http://www.roitt.com> › elspdf

PDF

"Immunological Accessory Molecules". In: Encyclopedia of ...



About featured snippets



Feedback

<https://medical-dictionary.thefreedictionary.com> › ...

definition of accessory molecules
by Medical dictionary

ac·ces·so·ry mol·e·cules ... surface receptors on
immune cells. which participate in cell-cell





what is Immunological synapse.



All

Images

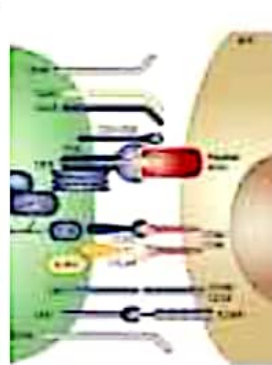
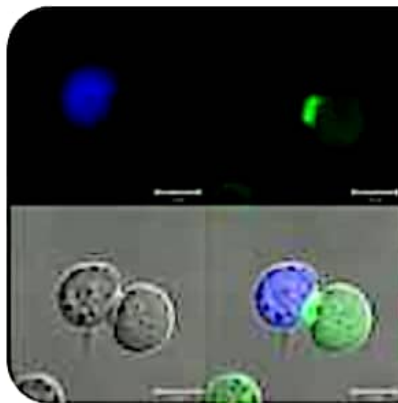
News

Videos

Books

Sea

Immunological synapse



In immunology, an immunological synapse is the interface between an antigen-presenting cell or target cell and a lymphocyte such as a T/B cell or Natural Killer cell. The interface was originally named after the neuronal synapse, with which it shares the main structural pattern. [Wikipedia](#)

Function



Importance



Cite

...

Abstract

The multi-chain T cell receptor/CD3 complex (TCR/CD3) plays a key role in antigen recognition, T cell activation and in consequence in triggering an antigen specific immune response. This process is induced by direct interaction of the TCR receptor with an antigen bound to the major histo-compatible complex on antigen-presenting cells. Upon the structural and functional cooperation of TCR receptor with CD3 complex, the activating signal is transmitted through the cell membrane to the nucleus. The pivotal role in signaling cascade plays CD3-zeta (zeta) chain, which triggers many biochemical events and second messenger activation, leading to the transcriptional factors expression and further T cell proliferation, effector function augmentation and cytokine production.

Similar articles



cell receptor/CD3 complex (TCR/CD3) plays a key role in antigen recognition, T cell activation and in...

<https://pubmed.ncbi.nlm.nih.gov> > ...

Structural and functional characteristics of the CD3 (T3)

by CD Tsoukas · 1987 · Cited by 14 — The CD3 (T3) molecular complex is noncovalently associated wit...

People also ask

What is the function of CD3?



The multi-chain T cell receptor/CD3 complex (TCR/CD3) plays a key role in **antigen recognition, T cell activation and in consequence in triggering an antigen specific immune response.**

 <https://pubmed.ncbi.nlm.nih.gov> > ...

[\[Structure and function of lymphocyte TCR/CD3 complex\] - PubMed](#)

More results

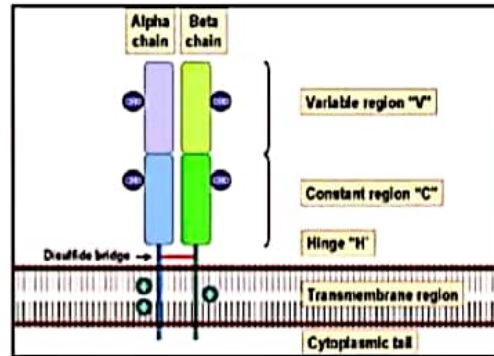
What is the major role of the CD3 complex which is associated with the TCR chains during T cell





Structure of the T cell Receptor

- Each T cell bears TCRs of only one specificity (allelic exclusion)



SlideShare

MHC and TCR

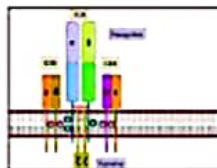
Visit

Images may be subject to copyright. [Learn More](#)

Related images

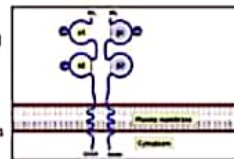
TCR and CD3 Complex

- TCR is closely associated with a group of 5 proteins collectively called the CD3 complex
 - γ chain
 - δ chain
 - ϵ chain
 - ζ chain
 - η chain
- CD3 proteins are invariant



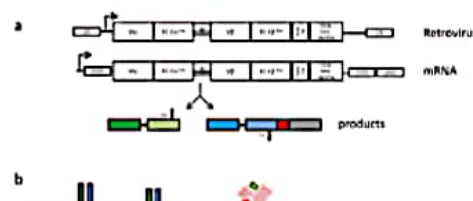
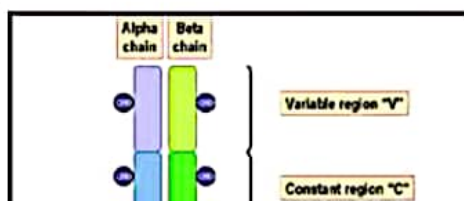
Structure of Class II MHC

- Four regions
 - Transmembrane region containing hydrophobic amino acids
 - A highly conserved $\alpha 2$ and a highly conserved $\beta 2$ domains to which CD4 binds
 - A highly polymorphic peptide binding region formed from the $\alpha 1$ and $\beta 1$ domains



MHC and TCR
slideshare.net

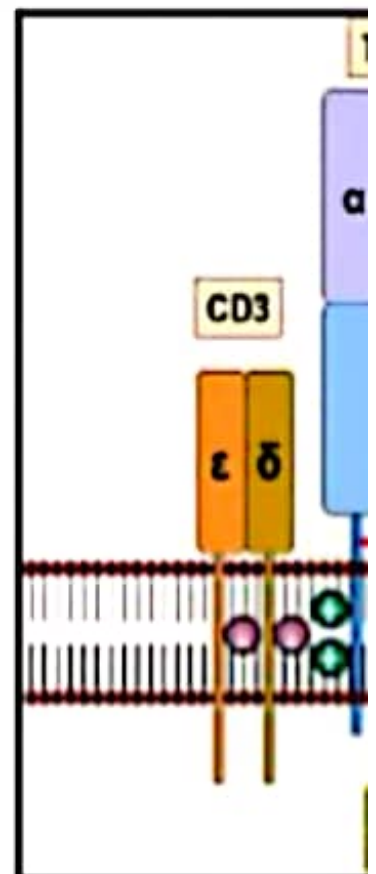
MHC and TCR
slideshare.net





TCR and CD3 Complex

- ▶ TCR is closely associated with a group of 5 proteins collectively called the CD3 complex
 - γ chain
 - δ chain
 - 2 ϵ chains
 - 2 ξ chains
- ▶ CD3 proteins are invariant





What is TCR?

- The T-cell receptor, or [TCR](#), is a molecule found on the surface of T cells, or T lymphocytes,[1] that is responsible for recognizing fragments of antigen as peptides bound to major histocompatibility complex (MHC) molecules. The binding between TCR and antigen peptides is of relatively low affinity and is degenerate: that is, many TCRs recognize the same antigen peptide and many antigen peptides are recognized by the same TCR.

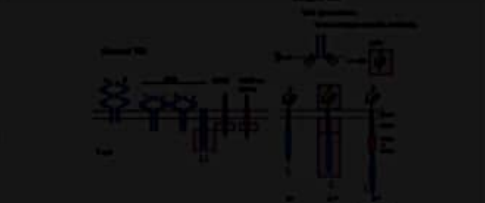
TCR is closely associated with a group of γ proteins collectively called the CD3 complex

- γ chain
- δ chain
- α chain
- β chain

CD3 proteins are invariant



MHC and TCR
slideshare.net



Structure of a classical T cell ...
researchgate.net



T-cell receptor - Wikipedia
en.wikipedia.org



CDR
pMHC binding
MHC contacts
Peptide contacts

T cell antigen recognition: Ev...
pnas.org